Pseudocode & Algorithm:

Pseudocode:

- *Input a set of 4 marks*
- *Calculate their average by summing and dividing by 4*
- *if average is below 50*

    Print "FAIL"

  else

    Print "PASS"

Pseudocode & Algorithm:

- Detailed Algorithm

```
Step 1:   Input M1,M2,M3,M4
Step 2:   GRADE ← (M1+M2+M3+M4)/4
Step 3:   if (GRADE < 50) then
              Print "FAIL"
          else
              Print "PASS"
          endif
```

The Flowchart:

- (Dictionary) A schematic representation of a sequence of operations, as in a manufacturing process or computer program.
- (Technical) A graphical representation of the sequence of operations in an information system or program. Information system flowcharts show how data flows from source documents through the computer to final distribution to users. Program flowcharts show the sequence of instructions in a single program or subroutine.
- Different symbols are used to draw each type of flowchart.

Example 2:

- Write an algorithm and draw a flowchart to convert the length in feet to centimeter.

Pseudocode:

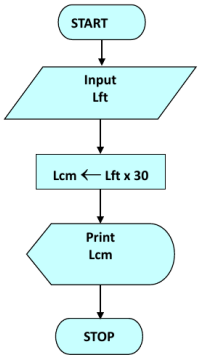
- *Input the length in feet (Lft)*
- *Calculate the length in cm (Lcm) by multiplying LFT with 30*
- *Print length in cm (LCM)*

Example 2:

Algorithm

- Step 1: Input Lft
- Step 2: Lcm ← Lft x 30
- Step 3: Print Lcm

Flowchart



Example 3:

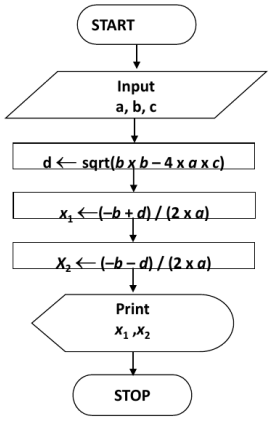
Write an algorithm and draw a flowchart that will read the two sides of a rectangle and calculate its area.

Pseudocode:

- *Input the width (W) and Length (L) of a rectangle*
- *Calculate the area (A) by multiplying L with W*
- *Print A*

Example 4:

- **Algorithm**:
- Step 1: Input a, b, c
- Step 2: $d \leftarrow \text{sqrt}(\ b \times b - 4 \times a \times c\ )$
- Step 3: $x1 \leftarrow (-b + d) / (2 \times a)$
- Step 4: $x2 \leftarrow (-b - d) / (2 \times a)$
- Step 5: Print $x1, x2$



Decision Structures:

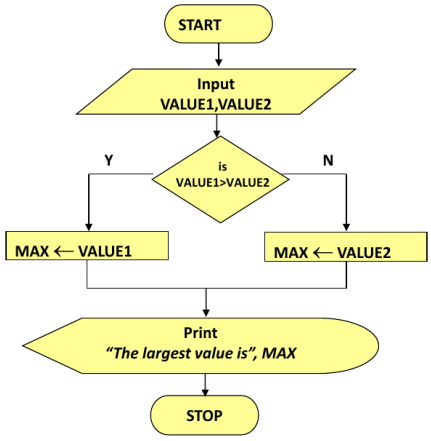
- The expression A>B is a logical expression
- *it describes a **condition** we want to test*
- ***if A>B is true (if A is greater than B)** we take the action on left*
- print the value of A
- ***if A>B is false (if A is not greater than B)** we take the action on right*
- print the value of B

IF–THEN–ELSE STRUCTURE:

- The structure is as follows

If (condition)then
true alternative
else
false alternative
endif

- **Example 5:**



NESTED IFS:

- One of the alternatives within an IF–THEN–ELSE statement
  - may involve further IF–THEN–ELSE statement

Example 6:

- Write an algorithm that reads **three** numbers and prints the value of the largest number.

Example 7:

Bonus Schedule	
OVERTIME – (2/3)*ABSENT	Bonus Paid
>40 hours >30 but ≤ 40 hours >20 but ≤ 30 hours >10 but ≤ 20 hours ≤ 10 hours	$50 $40 $30 $20 $10

Example 7:

Step 1: *Input* NAME,OVERTIME,ABSENT
Step 2: *if* (OVERTIME–(2/3)*ABSENT > 40) *then*
PAYMENT ← 50
else if (OVERTIME–(2/3)*ABSENT > 30) *then*
PAYMENT ← 40
else if (OVERTIME–(2/3)*ABSENT > 20) *then*
PAYMENT ← 30
else if (OVERTIME–(2/3)*ABSENT > 10) *then*
PAYMENT ←20
else
PAYMENT ← 10
endif
Step 3: *Print* "Bonus for", NAME "is $", PAYMENT

Example 7:

Flowchart: Draw the flowchart of the above algorithm?